CCD vs CMOS

The CCD (Charge Couple Device) is made up of small “photosites” arranged in patterns of groups and columns. This can range from 640x480 pixels to as high as 1600x1200 pixels. Together this makes the CCD, whose final pixel count ranges from a few hundred thousand to a few million pixels. The individual size of each pixel is about 10 millionth of a meter.

Transistors and sensors are located in the CMOS (Complementary Metal Oxide Semiconductor). These transistors are used to convert analog information into digital information, which is further processed by other circuitry and the image is formed. Due to this transistor the sensitivity of the CMOS is lower than that of the CCD since the light hitting the areas where transistors are located is not stored, and this information is completely lost. The CCD is more power consuming than the CMOS, with a normal CCD consuming 100 times more power. While CMOS is cheaper and easier to fabricate, it is more susceptible to noise and is one of the reasons why it can’t produce as high an image quality as the CCD. These are some of the prime reasons why CCDs are used in high-end, high-resolution cameras and lower end cameras, especially Web-cams, with resolutions of 640x480 have CMOS sensors.

on the speaker, but the image was dark where the intensity of light was kept low.

Resolution test

The resolution chart for this test had nine blocks, of which block number five and onwards were very small, with the ninth block being the most difficult to detect. The **Sony DCS-P1** passed this test with flying colours. This camera detected three-and-a-half blocks, which no other camera was able to detect, due to its 3.3-mega pixels CCD (Charge Couple Device) and the lens of the camera. Other cameras such as the Canon IXUS and Kodak DC-4800 could only detect three blocks. Since the distance between the cameras and the resolution chart was nearly 1 metre, it is understandable that the cameras were unable to detect at least half the total blocks. However, this test helps us understand which camera can pick up finer details if the target has sufficient light.

Feature test

We categorised the camera features as optics, operational modes and miscellaneous, of which optics and operational modes carry most weightage.

Optical/digital zooms: Optical zoom means the actual change in the focal length, which was in the range of 0x-3x in the cameras we tested. Digital zoom magnifies the image, but the quality of the image suffers.

Among the cameras tested, the Sony DSC-P1 had a 3x optical zoom and 6x digital zoom while the Kodak DC-4800 had a 3x optical and 2x digital zoom. Other cameras such as the Kodak DC-3400 and the Canon Power Shot had 3x optical and 2x digital zoom. The low-end cameras, which include the Intel Pocket Camera, D-Link Digi-Link and Kodak EZ200 came with a fixed focal length. These three cameras are VGA cameras, which don’t usually come with an adjustable lens.

The Intel and Digi-Link both gave tolerable photographs in bright light conditions. The Intel camera came with a lens adjustment for the video clip, which can be adjusted manually by rotating the plastic cap on the lens until the image looks clear enough on the computer monitor.

Sensing Element: There are two types of sensing elements used in digital cameras, CCD and CMOS. Out of the cameras in the 1024x768 pixels and above range, the Sony DSC-P1 camera had a maximum pixel count of 3.3 mega pixels while the Kodak DC-4800 had a 3.1 mega pixel count and was placed second. However, the Fuji, which produced very good image quality in terms of colour, had a CCD of 2.1 mega pixels. Good image quality is not attributed solely to the CCD but also the support electronics circuitry, which is where the Fuji scored higher.

Shutter speed: The faster the shutter speed, the crisper the picture. High-end cameras such as the Kodak DC-4800 and the Sony DSC-PI had a manual setting for the shutter speed, whereas cameras such as the Kodak DC-3400 and the Canon PowerShot A10 selected the shutter speed automatically, according to the exposure and other settings selected by the user. The Kodak EZ200 camera, which is a low-end camera, had the fastest shutter speed of 1/500 second and the D-link Digi-link had